

MATH 350 Advanced Calculus

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Outline

- 1 Real Analysis Lecture 2
 - Types of real numbers
 - Suprema and Infima

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Inductive sets

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Examples:

$$\mathbb{R}, \quad \mathbb{Q}, \quad (0, \infty), \quad \mathbb{Z}, \quad \mathbb{N}, \dots$$

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$$\mathbb{Q} = \{a/b : a, b \in \mathbb{Z}, b \neq 0\}.$$

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Now suppose $x \in S$. (This is our usual inductive assumption).

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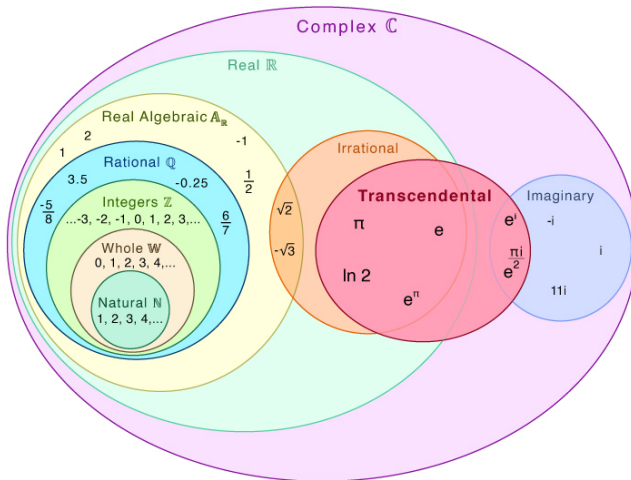
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Types of numbers

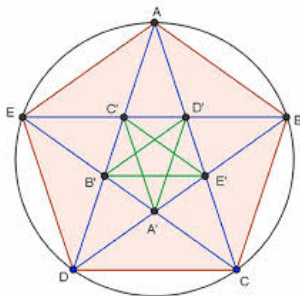
Transcendental Numbers



Irrational numbers

Numbers which are not of the form a/b with $a, b \in \mathbb{Z}$ are called **irrational**.

- Discovered by Hippasus, a pythagorean (a student of Pythagoras)



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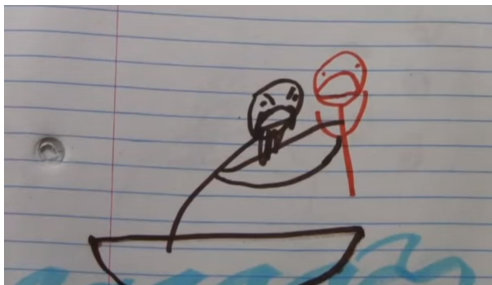
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- 3 transcendentals are mysterious ... but most real numbers are transcendental!

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- π is the supremum of

$$\{3, 3.1, 3.14, 3.141, 3.1415, 3.14159, 3.141592, \dots\}.$$

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Prove that if A is a set of integers that is bounded above, then A has a maximum.

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Therefore $k \leq 0$ and $a' \leq a$.

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It follows that a is an upper bound of A , and since $a \in A$ it is a maximum.

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In other words

an infimum is a **greatest lower bound**

Properties of Suprema

The first property of suprema is that they must be arbitrarily close to elements of the set.

Theorem (Approximation Property)

Let $S \subseteq \mathbb{R}$ be bounded above, and let $b = \sup(S)$. Then for all $\epsilon > 0$, there exists $a \in S$ with

$$b - \epsilon < a \leq b.$$

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Therefore there must exist $a \in S$ with $a > b - \epsilon$. □

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$$s_n \leq 1 + 1 + \frac{1}{2} + \frac{1}{2^2} + \frac{1}{2^3} + \cdots + \frac{1}{2^n} = 3 - \frac{1}{2^n} < 3$$

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so S is bounded above by 3. Therefore S has a supremum, $\sup(S)$.

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$$e^x = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \cdots$$

Important example: The number e

Even if ϵ is very small ($\epsilon = 0.000000001$), $\sup(S) - \epsilon$ is not an upper bound of S .

So there exists N with $s_N > \sup(S) - \epsilon$.

$$s_1 < s_2 < s_3 < s_4 < \cdots < \sup(S) - \epsilon < s_N < \cdots < \sup(S).$$

This says $\sup(S)$ is *really* close to s_N . Taking ϵ smaller and smaller

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We've just shown that $e^1 = e$ exists.

Properties of Suprema

Also, suprema play nicely with addition.

Theorem (Additive Property)

Let $A, B \subseteq \mathbb{R}$ be bounded above set

$$C = \{x + y : x \in A, y \in B\}.$$

Then C is bounded above and

$$\sup(C) = \sup(A) + \sup(B).$$

Properties of Suprema

Proof.

Let $a = \sup(A)$, $b = \sup(B)$, and $c = \sup(C)$.

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Thus $c - a$ is an upper bound of B , and it follows $b \leq c - a$.

Therefore $a + b \leq c$. □

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Use the completeness axiom to prove that the $\mathbb{Z}_+$ is not bounded above. (Apostol Theorem 1.17)

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Then $b - 1 < b$ by Axiom 7, so $b - 1$ cannot be an upper bound.

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It follows from Axiom 7 that $b < n + 1$.

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The number b is an upper bound and if $b' < b$, then b' is not.

Then $b - 1 < b$ by Axiom 7, so $b - 1$ cannot be an upper bound.

This means there exists $n \in \mathbb{Z}_+$ with $b - 1 < n$.

It follows from Axiom 7 that $b < n + 1$.

However, $n + 1 \in \mathbb{Z}$, so this contradicts b being an upper bound.

Archimedian property

Theorem (Apostol Theorem 1.18)

For every $x \in \mathbb{R}$, there exists $n \in \mathbb{Z}_+$ with $n > x$.

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If not, then x is an upper bound of $\mathbb{Z}_+$. □

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Theorem (Archimedean Property of Reals)

For every $x, y \in \mathbb{R}$ with $x > 0$, there exists $n \in \mathbb{Z}_+$ with $y < nx$.

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For every $x, y \in \mathbb{R}$ with $x > 0$, there exists $n \in \mathbb{Z}_+$ with $y < nx$.

Proof.

Replace x with y/x in the previous theorem. □